

VOLUME I - PROBLEM RESEARCH / DOCUMENT 1 OF 3

Problem Background

Monsoon Fog and Low-Visibility Haulage in the Bailadila Iron Ore Sector

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ABSTRACT

This document establishes the operational, meteorological, geotechnical and statutory context of the low-visibility haulage problem at NMDC's Bailadila complexes. It quantifies the production stake attached to the monsoon months, characterises the physics of orographic fog on the Bailadila ridge, describes the haul road geometry that converts poor visibility into a collision hazard, and frames the problem as a measurable loss of effective fleet hours rather than as a generic safety concern.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

Contents

1. Corporate and Production Context

- 1.1 NMDC in the national iron ore supply chain
- 1.2 The Bailadila production base
- 1.3 The 100 MT trajectory and why monsoon matters

2. The Physical Setting

- 2.1 Topography of the Bailadila range
- 2.2 Haul road network characteristics

3. Meteorology of the Problem

- 3.1 Monsoon regime, June to October
- 3.2 Orographic and advection fog formation
- 3.3 Visibility physics and the extinction coefficient

4. The Haulage System Under Stress

- 4.1 HEMM fleet and duty cycle
- 4.2 How visibility loss propagates into production loss
- 4.3 Quantifying the annual exposure

5. Safety and Accident Context

6. Statutory and Regulatory Frame

7. Problem Statement Restated in Engineering Terms

1. Corporate and Production Context

1.1 NMDC in the national iron ore supply chain

NMDC Limited, a Navratna public sector enterprise under the Ministry of Steel, is India's single largest iron ore producer. In FY 2025-26 the company recorded its highest-ever annual output of **53.15 million tonnes**, a year-on-year growth of approximately 21 per cent, becoming the first Indian producer to cross the 50 MT annual threshold. Sales in the same year reached approximately 50.23 MT. NMDC supplies close to one-fifth of the country's total iron ore requirement, which places the reliability of its production calendar squarely inside the national steel supply chain rather than inside a single company's operating plan.

The company's iron ore business is built on fully mechanised open cast complexes:

- ? **BIOM Kirandul Complex**, South Bastar, Dantewada district, Chhattisgarh - comprising Deposit-14 (commissioned 1968, India's first large-scale mechanised open cast iron ore mine), the 11C Project (14 NMZ + 11C part, commissioned 1988) and Deposit-11B (commissioned 2015).
- ? **BIOM Bacheli Complex**, Dantewada district, Chhattisgarh - working Deposits 5, 10 and 11A, with Deposit-4 recently commissioned.
- ? **Donimalai Complex**, Ballari district, Karnataka - Donimalai and Kumaraswamy mines.

Kirandul and Bacheli together constitute the Bailadila sector, which contributes on the order of **70 per cent of NMDC's total production**, or roughly 37 MTPA at present output levels.

1.2 The Bailadila production base

The Bailadila range holds some of the highest-grade haematite in the world, with typical run-of-mine Fe content in the 64-67 per cent band and low alumina and phosphorus. Mining is conducted by conventional shovel-dumper open cast methods on a hilltop and hill-slope deposit sequence, with ore transported from working benches down to crushing and screening plants, and thereafter evacuated primarily by the Kirandul-Kothavalasa (KK) rail line to Visakhapatnam, supplemented by an ore slurry pipeline to the Nagarnar steel plant.

Because the deposits sit on a ridge and the processing plants sit lower on the profile, the ore body is separated from the plant by a **descending haul road network with sustained gradients and switchbacks**. Every tonne of Bailadila ore passes through this network. There is no bypass, no alternate route, and no ability to store meaningful ore inventory on the bench. A constraint on haul road throughput is therefore a direct constraint on mine output.

1.3 The 100 MT trajectory and why monsoon matters

Aligned with the National Steel Policy 2017 target of 300 MT of domestic crude steel capacity by 2030-31 - which is projected to require 430-437 MT of iron ore annually - NMDC has committed to a production capacity of **over 100 MT by 2030**, with a near-term milestone of 60 MT in the current fiscal. Of the 100 MT, approximately **80 MT is expected from the Bailadila sector**.

The arithmetic of that ramp is unforgiving. Bailadila must roughly double its output from ~37 MTPA to ~80 MTPA. Capacity additions - new deposits, additional screening plant lines, expanded loading capability - address the annual average. They do not address the *distribution* of production across the year.

Roughly 42 per cent of the calendar year (June to October) falls inside the monsoon window. If effective haulage productivity during that window degrades by even 15-20 per cent relative to dry-season performance, the remaining seven months must absorb an increasingly severe catch-up load. At 37 MTPA this is a schedule pressure. At 80 MTPA, against a fleet and haul road network operating much closer to its geometric limits, it becomes a hard capacity ceiling.

This reframes the fog problem. It is not primarily a safety inconvenience to be managed by stopping work. It is a **capacity recovery problem** whose value scales linearly with the production ramp.

2. The Physical Setting

2.1 Topography of the Bailadila range

The Bailadila range is a north-south trending ridge in the Bastar craton rising to elevations in the region of 1,100-1,270 m above mean sea level, standing 400-700 m above the surrounding Indravati basin plateau. The name derives from the local Gondi description of the profile as resembling the hump of a bullock.

Three features of this topography matter directly to the problem:

- ? **Steep relief over short horizontal distance.** The ridge presents a pronounced windward face to the south-west monsoon flow, forcing rapid orographic ascent of moisture-laden air.
- ? **Working benches at or near cloud base.** During the monsoon the lifting condensation level frequently sits below the crest elevation. Hilltop benches are then physically inside cloud - not in fog that has drifted in, but in the condensation layer itself.
- ? **Vertical separation between mine and plant.** Ore descends several hundred metres of elevation between the highest working benches and the screening plants, which is what produces the long, graded, switchback haul profile.

2.2 Haul road network characteristics

Typical characteristics of a Bailadila-class hilltop haul road, indicative and to be confirmed against current mine plans:

Parameter	Typical value / range	Consequence for low visibility
Running gradient	1 in 16 maximum per DGMS practice; commonly 1 in 14 to 1 in 20 on ramps	Long braking distances on descent; loaded trucks cannot stop abruptly
Road width	3 to 3.5 x widest vehicle width for two-way traffic	Lateral margin for error is small relative to vehicle envelope
Curve radii	Tight switchbacks on ramp sections, often below 30 m	Line of sight around a bend can be shorter than stopping distance even in clear weather
Vertical profile	Repeated crest and sag curves following the ridge	Sight distance limited by geometry, compounded by fog
Edge protection	Berms of at least half the tyre height of the largest vehicle	The berm is the last line of defence, not a guidance aid
Traffic mix	Rear-dump trucks, water tankers, graders, dozers, service and light vehicles, personnel carriers	Mixed mass and speed profiles sharing one corridor
Surface	Compacted ore/waste fines, heavily watered in dry season, saturated in monsoon	Wet high-fines surfaces reduce available friction precisely when visibility is worst

The critical insight is that **the haul road is already at its geometric safety limit in clear weather**. Sight distance on a switchback descent is constrained by the curve, not by the atmosphere. Fog removes the remaining margin - and it removes it in the one place where a heavy loaded vehicle has the least ability to stop.

3. Meteorology of the Problem

3.1 Monsoon regime, June to October

The Bastar plateau receives the bulk of its precipitation from the south-west monsoon. Indicative regional climatology for the Dantewada / Bailadila area:

Month	Typical rainfall band	Relative humidity	Dominant visibility hazard
June	Onset; 150-250 mm	Rising, 80-90%	Onset squalls, first fog events
July	Peak; 350-500 mm	88-95%	Persistent hill cloud, heavy rain
August	Peak; 350-480 mm	88-95%	Persistent hill cloud, dense fog
September	Withdrawal phase; 200-300 mm	85-92%	Post-rain radiation/valley fog, low stratus
October	Retreat; 50-120 mm	75-88%	Early-morning radiation fog in valleys

Annual rainfall in the Bailadila belt is broadly in the 1,300-1,800 mm band, with the large majority concentrated in this five-month window. Wind on exposed ridge benches during active monsoon spells can be strong and gusty, which adds a second hazard: it displaces and reforms fog banks rapidly, so visibility is not merely low but **unstable on a timescale of seconds to minutes**.

3.2 Orographic and advection fog formation

Three mechanisms operate at Bailadila, often in combination:

Orographic cloud (hill fog)

Moisture-laden monsoon air is forced to ascend the windward face of the ridge. Adiabatic cooling drives the parcel to saturation, and condensation begins at the lifting condensation level. Where that level lies below the bench elevation, the bench is inside cloud. This is the dominant and most persistent mechanism during active monsoon spells and can hold for many continuous hours or days.

Advection fog

Warm, humid air moving over a cooler surface - for example, over rain-cooled rock and ore surfaces after a downpour - is chilled from below to its dew point. This mechanism produces the fast-moving, patchy fog banks characteristic of the post-rain period, where a truck can pass from 200 m visibility into 5 m visibility inside a single vehicle length.

Radiation fog

On clear, calm nights in the withdrawal phase (September-October), longwave radiative cooling of the ground chills the surface air layer to saturation. This is typically a valley-floor and pit-bottom phenomenon, strongest between the pre-dawn hours and mid-morning, and it dissipates with solar heating. It matters because it targets the night and early-morning shifts.

Operationally, these three mechanisms differ in a way that a solution must respect. Orographic cloud is **persistent and predictable** - it can be forecast hours ahead and planned around. Advection fog is **transient and abrupt** - it cannot be planned around and must be detected in real time. Radiation fog is **diurnal and localised** - it is predictable by time of day but spatially patchy within the pit. Any system that only handles one of the three will fail during a substantial fraction of the monsoon.

3.3 Visibility physics and the extinction coefficient

Meteorological Optical Range (MOR) is the distance at which the luminous flux of a collimated beam falls to 5 per cent of its original value. It relates to the atmospheric extinction coefficient by:

$MOR = 3.0 / \sigma_{ext}$ (Koschmieder relation, 5% contrast threshold)

where σ_{ext} is expressed per kilometre. The reported 3-5 m visibility on Bailadila hilltop benches therefore corresponds to an extinction coefficient on the order of **600-1000 per km** - an extraordinarily dense medium, at the upper end of what is encountered in natural fog anywhere.

The practical consequence is a matter of droplet physics. Fog attenuation is governed by Mie scattering, where the scattering efficiency peaks when the particle diameter is comparable to the wavelength of the incident radiation. Typical fog droplet diameters fall in the 1-20 micron range.

Sensing modality	Nominal wavelength	Relationship to fog droplet size	Expected behaviour at MOR 3-5 m
Human eye / visible camera	0.4-0.7 μm	Far smaller than droplets; strong Mie scattering	Effectively blind
NIR LiDAR (905 nm)	0.905 μm	Strong Mie scattering; heavy backscatter clutter	Severe degradation; range collapse
SWIR LiDAR (1550 nm)	1.55 μm	Slightly reduced but still strong scattering	Degraded; modest improvement only
LWIR thermal (8-14 μm)	8-14 μm	Comparable to largest droplets; reduced but non-trivial attenuation	Partial penetration; useful for warm targets at short-to-medium range
mmWave radar (77-79 GHz)	~3.9 mm	Three orders of magnitude larger than droplets; Rayleigh regime, minimal scattering	Substantially unaffected
UHF / sub-GHz RF ranging	0.1-1 m	Vastly larger than droplets	Unaffected
GNSS L-band	~0.19-0.24 m	Vastly larger than droplets	Unaffected by fog (other error sources dominate)

Table 3.1 - Why the choice of physical sensing modality is the first and most consequential design decision in this problem.

This table is the single most important technical finding in this volume. **No amount of image processing recovers information that never reached the sensor.** Visible-light enhancement - dehazing, contrast stretching, image fusion - operates on a signal that, at an extinction coefficient of 600 per km, has been attenuated beyond the noise floor within a few metres. The solution must be built on modalities whose wavelengths sit outside the Mie scattering peak for water droplets, with visible-band sensing relegated to a supporting role during partial-visibility conditions.

4. The Haulage System Under Stress

4.1 HEMM fleet and duty cycle

A Bailadila-class complex operates a mixed fleet typically including rear-dump trucks in the 35-100 t payload class, hydraulic and rope shovels and front-end loaders, dozers and graders for road maintenance, water tankers for dust suppression, drills, and a large population of light motor vehicles and personnel carriers. The light vehicle population is significant and, in collision terms, asymmetric: a light vehicle in the blind zone of a loaded rear-dump truck is the classic fatal interaction in open cast mining worldwide.

The haul cycle decomposes into: spotting and loading at the face, loaded haul, queuing at the crusher or dump, tipping, empty return, and queuing at the face. In clear conditions, loaded haul and empty return dominate the cycle. It is precisely these two segments that fog attacks.

4.2 How visibility loss propagates into production loss

The causal chain is direct and worth setting out explicitly, because it identifies where a technical intervention can be applied.

- ? **Stage 1 - Sight distance collapses.** MOR falls below the safe stopping distance for the prevailing speed and gradient.
- ? **Stage 2 - Operators self-limit.** Speed is reduced to restore the relationship between sight distance and stopping distance. Speed on a loaded descent may drop from a normal operating band to a crawl.
- ? **Stage 3 - Cycle time inflates.** Haul and return segments lengthen. Because these segments dominate the cycle, total cycle time rises materially.
- ? **Stage 4 - Fleet effectiveness falls.** With a longer cycle, the same truck count delivers fewer trips per shift. Shovel hang time increases as trucks arrive less frequently; shovel utilisation falls even though the shovel itself is unimpeded.
- ? **Stage 5 - Operations are suspended.** Below a threshold visibility, continued movement cannot be justified and haulage is stopped in whole or in part. Production during the stoppage is zero.
- ? **Stage 6 - Restart losses.** Resumption is not instantaneous. Operators must be recalled, the fleet re-sequenced, queues cleared. Each stop-start cycle carries a fixed overhead independent of the length of the stoppage.

Stage 6 is routinely underestimated. In a regime of transient advection fog, a shift may contain several short stoppages. The sum of the restart overheads can exceed the sum of the stoppage durations themselves. A system that reduces the *number* of stoppages by riding through short fog banks therefore delivers benefit out of proportion to the visibility improvement it provides.

4.3 Quantifying the annual exposure

The following framework is offered as a structure for quantification rather than as a claim of measured values. Actual figures must be derived from NMDC dispatch logs, fleet management system records and shift reports.

Let:

- ? H_m = total scheduled haulage hours across the monsoon window (June-October)
- ? f_{fog} = fraction of H_m during which MOR falls below the operational threshold
- ? d_{speed} = mean proportional reduction in haul speed during fog-affected hours that are still worked
- ? f_{stop} = fraction of H_m lost to full suspension
- ? $n_{restart}$ = number of discrete stop-start events, each carrying overhead t_r

Effective lost haulage capacity, expressed as equivalent lost hours, is approximately:

$$L = H_m [f_{stop} + (f_{fog} - f_{stop}) d_{speed}] + n_{restart} * t_r$$

Converting to tonnes requires the mean haulage productivity in tonnes per effective fleet hour. The point of setting the model out in this form is that it isolates **four independent levers**, each of which a technology intervention can address separately:

Lever	Physical meaning	Which capability addresses it
Reduce f_{stop}	Fewer hours of full suspension	Guidance and collision avoidance that make movement defensible below the current visual threshold
Reduce d_{speed}	Higher speeds retained during fog	Confident lane-keeping guidance and forward hazard detection
Reduce n_{restart}	Fewer discrete stop-start events	Short-horizon fog nowcasting and graded operating modes rather than binary go/no-go
Reduce t_r	Faster recovery per event	Centralised fleet sequencing and automated re-dispatch from the control room

A solution that addresses only collision avoidance touches the first lever and part of the second. A solution that addresses only forecasting touches the third. The full benefit case requires all four, which is the principal argument for an integrated system rather than a point product.

5. Safety and Accident Context

Transport and tramming machinery is the dominant cause of fatal and serious accidents in Indian open cast mines. The Directorate General of Mines Safety issued Technical Circular No. 03 of 2024 specifically addressing safety in transport and tramming machinery operations at opencast mines, in response to a cluster of vehicle-related accidents at Indian mining sites.

The accident archetypes that low visibility makes materially more likely:

- ? **Rear-end collision, loaded on descent.** The classic and most consequential. A following truck closes on a stopped or slowed lead truck inside its sight distance. Energy is high; the gradient works against the following vehicle.
- ? **Run-over of light vehicle or person.** A light vehicle or pedestrian in the blind zone of a rear-dump truck. Fog compounds an already severe blind-spot geometry.
- ? **Edge departure.** Loss of lateral reference on a ramp leading to contact with or departure over the berm. The berm is designed to arrest a vehicle, but a berm strike at speed can still overturn a truck.
- ? **Head-on on two-way sections.** Where the road is narrow and a vehicle drifts from its side of the running surface.
- ? **Collision with stationary plant or parked equipment.** Graders, water tankers and service vehicles occupying part of the running surface are effectively invisible in dense fog.
- ? **Tipping-area incidents.** Dump edge, spotting and reversing operations near a crest, where a positional error has severe consequences.

The operational reality is that the current control for all of these in dense fog is a single administrative measure: stop working. That control is effective for safety and catastrophic for production. The design objective of this research programme is to introduce engineering controls that make a graded, continuous operating response possible in place of a binary one.

6. Statutory and Regulatory Frame

Any proposed solution operates inside a defined regulatory envelope. Relevant instruments include:

- ? **The Mines Act, 1952** and the **Metalliferous Mines Regulations, 1961 (MMR 1961)** - the governing statute and regulations for iron ore mining. Responsibility under the Act attaches to both the lease holder and any contractor deploying HEMM on the lease.
- ? **DGMS Technical Circular No. 03 of 2024** - safety of transport and tramming machinery in opencast mines. Mandates daily pre-operational checklists for all TTM, requires rear-view cameras and Proximity Detection Systems on dumpers, sets haul road gradient limits at 1 in 16, and requires certified operators with refresher training.
- ? **DGMS Technical Circular No. 06 of 2020** - comprehensive safety features for HEMM and heavy/light vehicles in open cast mines, covering rear vision cameras, operator fatigue warning, auto-dipping, collision avoidance, automatic fire detection and suppression, dump body raised position indicators, and proximity warning devices that detect static and moving objects and scale the warning with speed and direction.
- ? **DGMS Circular 08/2017** and Regulation 107/108 provisions on audio-visual alarms for surface transport machinery and HEMM.
- ? Requirements on haul road design, berm height, gradient, width, signage, drainage and illumination.

Two implications follow. First, **the regulatory direction of travel already favours this class of solution** - proximity detection and collision avoidance are not novel asks but existing mandates. Second, **any intervention that alters vehicle control authority** - automatic braking, speed limiting, or any form of autonomy - requires engagement with DGMS on approval, testing and permissioning. The design should therefore separate advisory functions, which sit comfortably inside current practice, from intervention functions, which require a defined approval pathway.

7. Problem Statement Restated in Engineering Terms

Consolidating the analysis, the problem can be stated with the precision needed to drive design:

Problem. For approximately five months a year, atmospheric extinction on Bailadila hilltop haul roads intermittently rises to 600-1000 per km, collapsing meteorological optical range to 3-5 m. This is below the safe stopping distance of a loaded rear-dump truck on a 1-in-16 descent at any useful speed. In the absence of a non-optical means of establishing (a) the vehicle's lateral position within the running surface, (b) the range and closing rate of forward hazards, and (c) the position and intent of all other traffic in the corridor, the only available control is administrative suspension of haulage. The resulting loss of effective fleet hours, together with the fixed overhead of repeated stop-start cycles, is a material constraint on monsoon production - and becomes a binding constraint on the Bailadila sector's path from approximately 37 MTPA to a targeted 80 MTPA by 2030.

The corresponding solution requirement:

- ? Establish vehicle position on the haul road to sub-metre lateral accuracy using means independent of optical visibility.
- ? Detect and range forward and lateral hazards, both cooperative and non-cooperative, using modalities operating outside the Mie scattering peak for water droplets.
- ? Share position, velocity and intent between all vehicles in the corridor and with a central authority, over a link that is itself unaffected by fog.
- ? Present the resulting situational picture to the operator in a form that is usable at a glance, under vibration, in a high-noise cab, by an operator who cannot see out of the windscreen.
- ? Provide the control room with a real-time common operating picture, a short-horizon visibility forecast, and the means to sequence and re-dispatch the fleet through graded operating modes.
- ? Do all of this within the statutory envelope of the Mines Act, MMR 1961 and the relevant DGMS circulars, and at a cost and ruggedness appropriate to retrofit across a large mixed fleet.

The three documents of Volume I develop this statement. Document 2 details the specific failure modes of existing controls. Document 3 converts the requirement into a structured, testable specification.